

Math 150A Section 6
Fall 2020
Exam I
September 29, 2020
Time Limit: 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 7 pages (including this cover page) and 6 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam. However, you may use a single, handwritten, one-sided notesheet and a *basic* calculator.

You are required to show your work on each problem on this exam, unless otherwise stated. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.
- Unless otherwise stated, all angles are in **radians**

Problem	Points	Score
1	10	
2	10	
3	5	
4	10	
5	10	
6	5	
Total:	50	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** If the statement is true, write TRUE (all caps, all spelled out). Otherwise write FALSE (all caps, all spelled out).

(a) the function $\sin(x^2)$ is even

(b) the limit $\lim_{x \rightarrow \infty} f(x)$ is always either 0 or ∞

(c) for any functions $f(x)$ and $g(x)$ with $\lim_{x \rightarrow 0} f(x) = 0$, the limit $\lim_{x \rightarrow 0} [f(x)g(x)] = 0$.

(d) a rational function is continuous at every point in its domain

(e) if $f(x)$ has a discontinuity at 3, then $f'(3)$ does not exist

2. (10 points) Calculate the following. Carefully show your work.

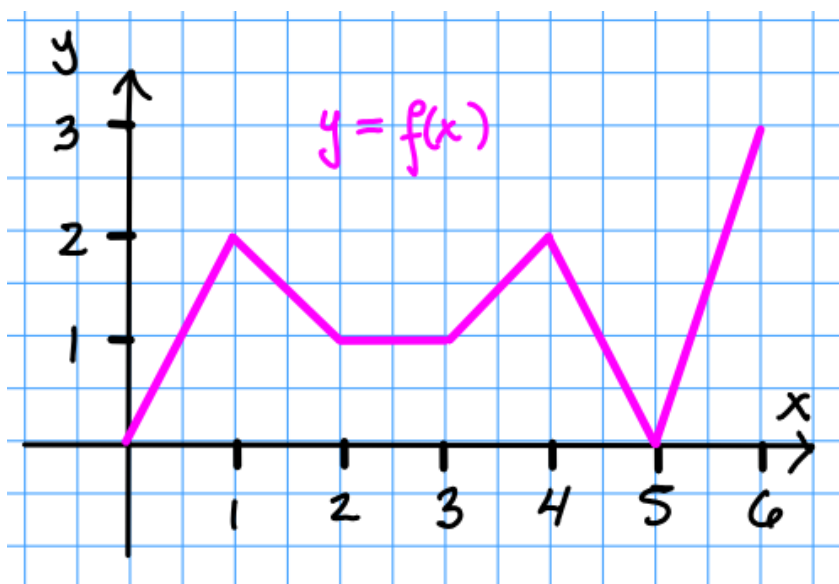
(a) $\lim_{x \rightarrow 2} \frac{\sqrt{x+2} - \sqrt{2x}}{x-2}$

(b) $\lim_{x \rightarrow 1} \frac{\frac{x+1}{x} - \frac{2}{x}}{x-1}$

(c) $f'(x)$ for $f(x) = x \sin(x)$

(d) $g'(x)$ for $g(x) = \tan\left(\frac{1+x}{1-x}\right)$

3. (5 points) Draw a graph of the derivative of the function $f(x)$ whose graph is given below. Carefully label the x and y axes and also indicate any points where the derivative does not exist.



4. (10 points) In lectures, we went over many important examples. For each of the following, give an explicit example. You do not need to show any work.

(a) An odd function.

(b) A limit which does not exist

(c) A function with a jump discontinuity.

(d) A function with a vertical asymptote at $x = 2$

(e) A function which is continuous at 0 but which is not differentiable at 0.

5. (10 points) Consider functions $f(x)$, $g(x)$, and $h(x)$ whose values satisfy

x	1	2	3	4	5	6	7
$f(x)$	1	1	2	3	5	8	13
$f'(x)$	0	1	3	2	7	9	4

x	1	2	3	4	5	6	7
$g(x)$	8	6	7	5	3	0	9
$g'(x)$	0	1	0	1	2	3	5

x	1	2	3	4	5	6	7
$h(x)$	2	3	5	7	1	13	17
$h'(x)$	4	3	2	1	2	3	4

Evaluate the following expressions.

(a) $F'(5)$ for $F(x) = f(x) + 2g(x) - h(x)$

(b) $F'(3)$ for $F(x) = xg(x)$

(c) $F'(1)$ for $F(x) = f(x)g(x)h(x)$

(d) $F'(2)$ for $F(x) = f(x^2)$

(e) $F'(4)$ for $F(x) = \frac{g(x)}{x^2 f(x)}$

6. (5 points) Using **only** the formal definition of a derivative and any limit laws, calculate the derivative of

$$f(x) = \frac{1}{x}.$$